Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/31/2025
PI Name:	Shelhamer, Mark Sc.D.		
Project Title:	Investigation of Partial-g Effects on Ocular Alignment		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Baltimore	**State:**	MD
Zip Code:	21205-1832	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019 HERO 80JSC019N0001-FLAGSHIP & OMNIBUS: Human Research Program Crew Health. Appendix A&B
Start Date:	07/28/2020	**End Date:**	07/27/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 07/27/2025 per NSSC information (Ed., 12/30/24) NOTE: End date changed to 07/27/2024 per NSSC information (Ed., 3/20/23)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schubert, Michael Ph.D. (Johns Hopkins University)		
Grant/Contract No.:	80NSSC20K1498		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project will provide information on binocular alignment as a measure of otolith asymmetry – more specifically as a measure of the neural compensation for asymmetry, which changes as a function of g level. This low-level function is easily and rapidly measured, and has been validated in vestibular patients and parabolic flight. The project draws on related Human Research Program (HRP) initiatives: Sensorimotor Assessment and Rehabilitation Apparatus (NNX10AO19G, 2010-2014) and Assessment of Otolith Function and Asymmetry as a Corollary to Critical Sensorimotor Performance in Missions of Various Durations (80NSSC19K0487, 2019-2027). Based on our previous studies, we anticipate a threshold of about 0.3 g, where there is a transition from ocular alignment that prevails in 1 g to that which is normal in 0 g (Karmali et al., J Vestibular Res 16:117-125, 2006). A subsequent model suggests a slightly higher (but not abrupt) transition at about 0.6 g (Beaton et al., Frontiers Syst Neurosci 9, 2015); thus, we predict a switching threshold in the range of 0.3 to 0.6 g.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ability to evaluate vestibular (otolith) function, via simple and rapid testing with minimal apparatus, is useful in a variety of clinical settings. Clinical evaluation of our device and procedures is ongoing, for application to vestibular patients and military personnel with blast injuries.
Task Progress:	Our alignment testing was implemented on a Meta Quest 2 virtual reality headset. An Xbox hand controller was used by a subject to interact with the headset; modifications were made to the controllers to remove unnecessary (and distracting) controls in order to simplify operation in flight. Extensive validation, verification, and familiarization were performed before flight. (A single parabolic flight with Zero G Corporation, to verify the operation of the headset in altered gravity, was twice scheduled and canceled.) Flights took place in the summer of 2023 in Bordeaux, France. Logistics, planning, and documentation for parabolic flights were carried out in association with Novespace, which provided test subjects. These subjects were shared with Gary Strangman's project. Subjects had previous parabolic-flight experience. This partly mimics the spaceflight situation that is relevant to this project: astronauts on their way to the Moon (Mars), who have adapted to 0g during transit, and who are then subjected to altered g during and after lunar (Martian) landing. Twelve subjects were tested at four g levels in parabolic flight: 0, 0.25, 0.50, 0.75. The first flight was dedicated solely to 0g parabolas. In the other three flights, the three other g-levels were presented, in a different order on each flight Baseline measures were made at 1g. Vertical and torsional alignment were obtained for each subject at each g level. There is an abrupt (rather than smooth) transition in alignment as g level increases. This alignment occurs, in some cases, between 0 and 0.25g. It is unclear if this occurs at a g level less than or greater than 0.16g (lunar). Statistical testing verified these indications of a low-g transition (between 0g and 0.25g) – not a smooth change – as a function of g level. This was seen in six subjects in vertical alignment and ten subjects in torsional alignment. Six TAN subjects and three VAN subjects satisfy a statistical criterion for this low-g transition, which was performed as follows: • remove outliers at each g level (more than three scaled median absolute deviations from the median) • test for no difference between medians at 0.25 and 0.5 g (Kruskal-Wallis, $p>0.1$) – AND – • test for difference in medians between 0g and grouped 0.25 & 0.5 g (Kruskal-Wallis, $p<0.1$). In some cases, there may be an additional abrupt transition in alignment between the higher g levels (0.5 & 0.75) and 1g. This implies that partial-g is different from both 0g and 1g, with a low-g transition and a high-g transition. This may reflect the novelty of hypo-g relative to lifelong experience at 1g. A significant concern was stability of the visual images on the VR display under altered gravity levels. We thought that this might be adversely impacted if the system uses inertial sensors (accelerometers) to maintain stability of the rendered visual scene as the head moves. Efforts to resolve this were aided by discussions with Laurence Harris and Pascal Serrarens, who have experience with the Quest platform in parabolic flight. Attempts to perform a validation flight with Zero G Corporation were unsuccessful. In flight, there was, in fact, initial drift of the visual image during the first few seconds of each transition into a new g level. This resolved quickly, with the image returning to the desired screen location and remaining stable. Operationally this was accommodated by having subjects ignore the initial few seconds of each transition, until the image was stable.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Conference Materials (Downloadable)	Shelhamer M, Schubert MC. "Spaceflight implications of otolith asymmetry. " 2025 Vestibular-Oriented Research Meeting, Boulder CO, May 19-22, 2025. , May-2025 VOR meeting abstract 2025.pdf